

The Very Basic Note in Geometry

Version 1.5.1

15-7-2023

Contents

1	Angles	2
2	Congruent Triangles and Similar Triangles	4
3	Circles	7
4	Trigonometry	13
5	Triangle Geometry	17
5.1	General Facts	17
5.2	Triangle Centres	21
6	Quadrilaterals	28
	Links	31

1 Angles

Proposition 1.1. The sum of all angles around a point is 360° .

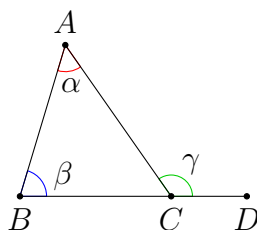


For example, in the left figure, we have $\alpha + \beta + \gamma = 360^\circ$. As a corollary, the sum of all angles on one side of a straight line is 180° . Indeed, in the right figure, ABC is a straight line if and only if $\alpha + \beta = 180^\circ$.

Proposition 1.2. Let $n \geq 3$ be an integer. Then the sum of all internal angles of any n -gon is $(n - 2) \times 180^\circ$.

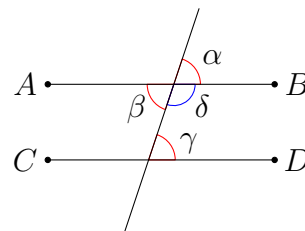
In particular, the sum of all internal angles of a triangle is 180° .

Proposition 1.3. In $\triangle ABC$, let D be a point on the extension of BC beyond C . Label the angles as shown. Then $\gamma = \alpha + \beta$.



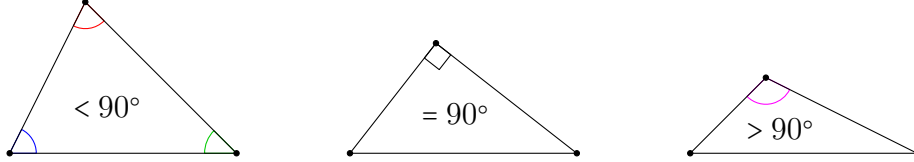
Proposition 1.4. As shown in the figure, suppose $AB \parallel CD$. Then the following hold.

- (a) $\alpha = \beta$
- (b) $\alpha = \gamma$
- (c) $\beta = \gamma$
- (d) $\gamma + \delta = 180^\circ$

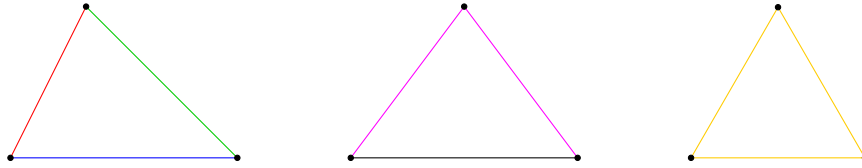


Conversely, each of (b), (c) and (d) implies $AB \parallel CD$.

Definition 1.1. Let $\triangle ABC$ be a triangle. We say that $\triangle ABC$ is an *acute triangle*(銳角三角形) if all interior angles are less than 90° . We say that $\triangle ABC$ is a *right-angled triangle*(直角三角形) if one of the interior angles is 90° . We say that $\triangle ABC$ is an *obtuse triangle*(鈍角三角形) if one of the interior angles is larger than 90° .



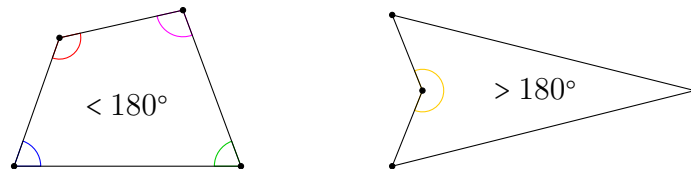
Definition 1.2. Let $\triangle ABC$ be a triangle. We say that $\triangle ABC$ is a *scalene triangle*(不等邊三角形) if all sides have different lengths. We say that $\triangle ABC$ is an *isosceles triangle*(等腰三角形) if at least two sides have the same length. We say that $\triangle ABC$ is an *equilateral triangle*(等邊三角形) if all sides have the same length.



Proposition 1.5. Let $\triangle ABC$ be a triangle. Then $AB = AC$ if and only if $\angle B = \angle C$.

Indeed, there is a stronger result that $AB > AC$ if and only if $\angle C > \angle B$. As a corollary, all interior angles of an equilateral triangle are 60° .

Definition 1.3. We say that a polygon is a *convex polygon*(凸多邊形) if all interior angles are less than 180° . We say that a polygon is a *concave polygon*(凹多邊形) if at least one interior angle is larger than 180° .



2 Congruent Triangles and Similar Triangles

Definition 2.1. $\triangle ABC$ and $\triangle XYZ$ are said to be *congruent*(全等) if their corresponding sides and corresponding angles are equal, i.e.

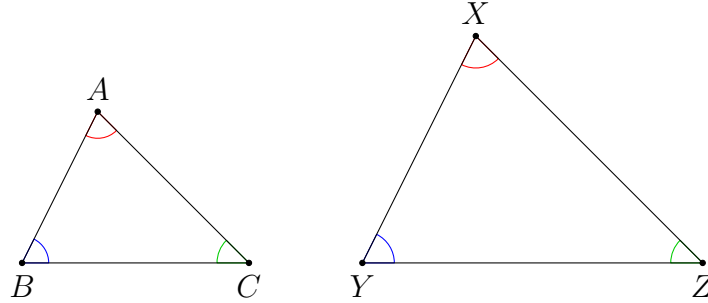
$$\begin{aligned}\angle A &= \angle X, & \angle B &= \angle Y, & \angle C &= \angle Z, \\ BC &= YZ, & CA &= ZX, & AB &= XY.\end{aligned}$$

In that case, we write $\triangle ABC \cong \triangle XYZ$.

Definition 2.2. $\triangle ABC$ and $\triangle XYZ$ are said to be *similar*(相似) if their corresponding angles are equal and their corresponding sides are in fixed proportion, i.e.

$$\begin{aligned}\angle A &= \angle X, & \angle B &= \angle Y, & \angle C &= \angle Z, \\ \frac{AB}{XY} &= \frac{BC}{YZ} = \frac{CA}{ZX}.\end{aligned}$$

In that case, we write $\triangle ABC \sim \triangle XYZ$.



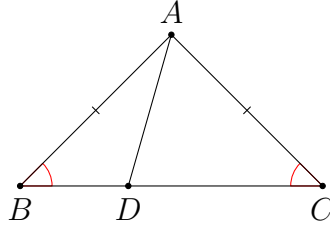
When using the notations $\triangle ABC \cong \triangle XYZ$ or $\triangle ABC \sim \triangle XYZ$, we must write in accordance with the order of the congruency or similarity. For example, in this case, vertex A in $\triangle ABC$ corresponds to vertex X in $\triangle XYZ$.

Proposition 2.1. Let $\triangle ABC$ and $\triangle XYZ$ be any triangles. Then the following are equivalent.

- (i) $\triangle ABC \cong \triangle XYZ$
- (ii) (SSS) $AB = XY, BC = YZ, CA = ZX$
- (iii) (SAS) $AB = XY, \angle B = \angle Y, BC = YZ$
- (iv) (RHS) $\angle A = \angle X = 90^\circ, AB = XY, BC = YZ$
- (v) (AAS) $\angle A = \angle X, \angle B = \angle Y, BC = YZ$
- (vi) (ASA) $\angle A = \angle X, AB = XY, \angle B = \angle Y$

For (iii), the pair of equal angles must be the included angles between the equal sides. For example, even if we have $AB = XY$, $BC = YZ$ and $\angle C = \angle Z$, it is not guaranteed that $\triangle ABC \cong \triangle XYZ$ since $\angle C$ and $\angle Z$ are not the included angles.

Example 2.1. Let $\triangle ABC$ be an isosceles triangle with $AB = AC$. Let D be a point on the side BC . Note that $AB = AC$, $AD = AD$ and $\angle ABD = \angle ACD$. But then $\triangle ABD$ and $\triangle ACD$ need not be congruent.

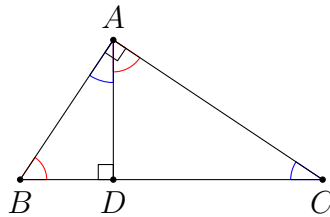


Proposition 2.2. Let $\triangle ABC$ and $\triangle XYZ$ be any triangles. Then the following are equivalent.

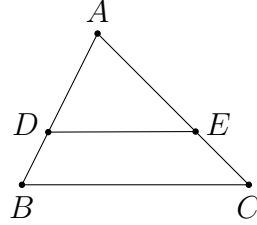
- (i) $\triangle ABC \sim \triangle XYZ$
- (ii) (3 sides proportional) $\frac{AB}{XY} = \frac{BC}{YZ} = \frac{CA}{ZX}$
- (iii) (2 sides proportional, included angles) $\frac{AB}{XY} = \frac{BC}{YZ}$, $\angle B = \angle Y$
- (iv) (equiangular) $\angle A = \angle X$, $\angle B = \angle Y$, $\angle C = \angle Z$

As in the case of congruent triangles, it is necessary for (iii) to be the included angles. For (iv), note that since the sum of internal angles of a triangle is fixed, it is sufficient to have two pairs of equal angles instead of three pairs.

Example 2.2. In a right-angled $\triangle ABC$, $\angle BAC = 90^\circ$. Let D be the foot of altitude from A . Then we have $\triangle DBA \sim \triangle ABC \sim \triangle DAC$. Indeed, we easily find that $\angle DBA = 90^\circ - \angle BAD = \angle DAC$ and similarly $\angle BAD = \angle ACD$.



Theorem 2.1. (Intercept theorem(截線定理)) In $\triangle ABC$, let D and E be points on the sides AB and AC respectively. Then $DE \parallel BC$ if and only if $\frac{AD}{DB} = \frac{AE}{EC}$.



Proof. If $\frac{AD}{DB} = \frac{AE}{EC}$, then $\frac{AD}{AB} = \frac{AE}{AC}$. Together with $\angle DAE = \angle BAC$, we have $\triangle DAE \sim \triangle BAC$ by [proposition 2.2\(iii\)](#). Thus, $\angle ADE = \angle ABC$, and $DE \parallel BC$ by the converse of [proposition 1.4\(b\)](#).

Conversely, if $DE \parallel BC$, then we can apply [proposition 1.4\(b\)](#) to get $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$. Thus, $\triangle ADE \sim \triangle ABC$ by [proposition 2.2\(iv\)](#). This gives $\frac{AD}{AB} = \frac{AE}{AC}$, and hence $\frac{AD}{DB} = \frac{AE}{EC}$. $\square$

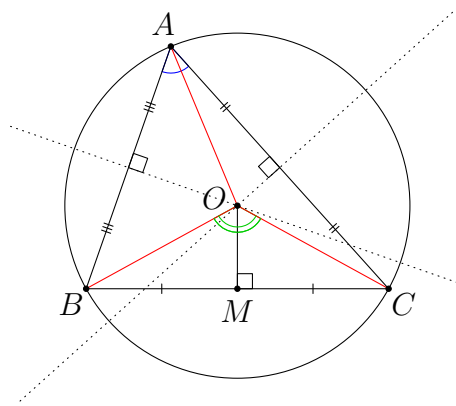
In view of the proof, whenever there is a pair of parallel lines, we can always find some similar triangles.

3 Circles

Proposition 3.1. For any three points not lying on a straight line, there exists a unique circle passing through all three points.

Proof. Consider three points A, B, C not lying on a straight line. If we wish to find a circle with centre O passing through A, B, C , we must have $OA = OB = OC$. Conversely, whenever this holds, we can always draw such a circle uniquely.

Note that $OA = OB$ if and only if O lies on the perpendicular bisector of AB (which can be proved using congruent triangles). Similarly, $OA = OC$ if and only if O lies on the perpendicular bisector of AC . Since AB and AC are not parallel, these perpendicular bisectors have a unique intersection point. This is exactly the desired point O . $\square$



Proposition 3.2. Let $\triangle ABC$ be an acute triangle. Let O be the centre of the circle passing through A, B, C , and let M be the midpoint of BC . Then the following hold.

- (a) $\angle BOC = 2\angle BAC$
- (b) $OM \perp BC$

Proof.

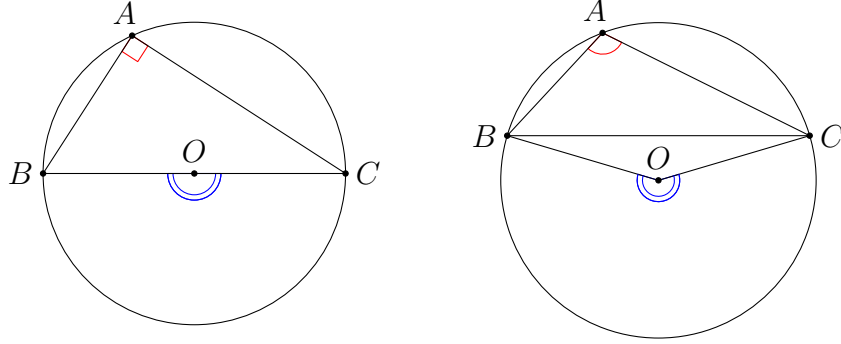
- (a) Note that $\angle OBA = \angle OAB$ and $\angle OCA = \angle OAC$ by [proposition 1.5](#). Using [proposition 1.3](#), we have

$$\angle BOC = \angle OBA + \angle OAB + \angle OAC + \angle OCA = 2\angle OAB + 2\angle OAC = 2\angle BAC.$$

- (b) This follows from the fact that O lies on the perpendicular bisector of BC (see the proof of [proposition 3.1](#)).

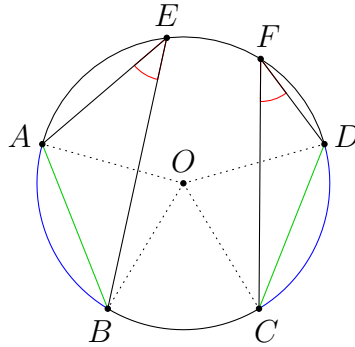
$\square$

Proposition 3.2 also holds if $\triangle ABC$ is not acute. In particular, if $\angle BAC = 90^\circ$, then $\angle BOC = 180^\circ$, and hence BC is a diameter of the circle. Conversely, if BC is a diameter of a circle, then $\angle BAC = 90^\circ$ for any point A lying on the circle. If $\angle BAC > 90^\circ$, then $\angle BOC$ refers to the reflex angle larger than 180° .



Proposition 3.3. Let A, B, C, D be points on a circle. Let E be a point on the major $\widehat{AB}$, and let F be a point on the major $\widehat{CD}$. Then the following are equivalent.

- (i) $\angle AEB = \angle CFD$
- (ii) $\widehat{AB} = \widehat{CD}$ (both are minor arcs or both are major arcs)
- (iii) $AB = CD$



Proof. Let O be the centre of the circle. We claim that all conditions are equivalent to (iv) $\angle AOB = \angle COD$.

- (i) $\Leftrightarrow$ (iv) since $\angle AOB = 2\angle AEB$ and $\angle COD = 2\angle CFD$ by **proposition 3.2(a)**.
- (ii) $\Leftrightarrow$ (iv) since the arc length is proportional to the angle at centre.
- (iii) $\Leftrightarrow$ (iv) since both are equivalent to $\triangle AOB \cong \triangle COD$ (noting $OA = OC$ and $OB = OD$).

□

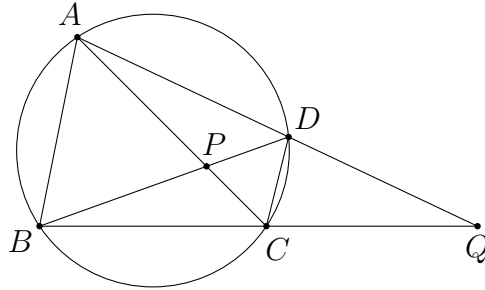
The result also holds if E and F lie on minor $\widehat{AB}$ and minor $\widehat{CD}$ respectively.

Next, as a corollary of [proposition 3.1](#), it is obvious that not every group of four points could lie on the same circle. This leads to the following definition.

Definition 3.1. Four points are said to be *concylic* (共圓) if there exists a circle passing through all four points. A quadrilateral $ABCD$ is called a *cyclic quadrilateral* (圓內接四邊形) if A, B, C, D are concyclic.

Proposition 3.4. Let $ABCD$ be a convex quadrilateral. Suppose the diagonals AC and BD meet at P , the ray AD and the ray BC meet at Q . Then the following are equivalent.

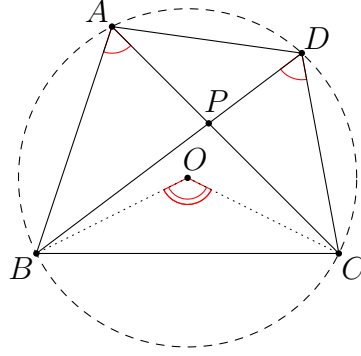
- (i) A, B, C, D are concyclic
- (ii) $\angle BAC = \angle BDC$
- (iii) $\angle BAD = \angle QCD$
- (iv) $\angle BAD + \angle DCB = 180^\circ$
- (v) $PA \times PC = PB \times PD$
- (vi) $QD \times QA = QC \times QB$



Proof. We prove the equivalence in a few steps.

- (i) $\Leftrightarrow$ (ii): Let O_1 and O_2 be the centres of the circles passing through A, B, C and D, B, C respectively. Note that both of them lie on the perpendicular bisectors of BC . Therefore, $O_1 = O_2$ if and only if $\angle BO_1C = \angle BO_2C$. Now, (i) is equivalent to $O_1 = O_2$, while (ii) is equivalent to $\angle BO_1C = \angle BO_2C$ by [proposition 3.2\(a\)](#). This proves (i) $\Leftrightarrow$ (ii).
- (ii) $\Leftrightarrow$ (v): If (ii) holds, together with $\angle APB = \angle DPC$, we can find that $\triangle PAB \sim \triangle PDC$. This implies $\frac{PA}{PD} = \frac{PB}{PC}$, and hence (v).

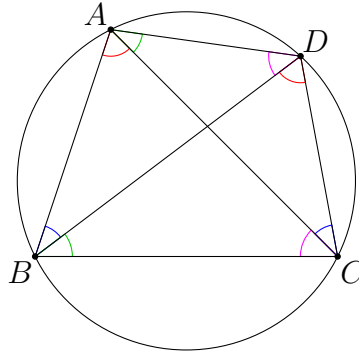
Conversely, if (v) holds, then we have $\triangle PAB \sim \triangle PDC$. This implies (ii).



- (i) $\Leftrightarrow$ (vi): This is similar to the above proof. Note that (i) is also equivalent to $\angle QBD = \angle QAC$. Both (i) and (vi) are the same as $\triangle QCA \sim \triangle QDB$.
- (iii) $\Leftrightarrow$ (vi): Again the proof is similar. Both are equivalent to $\triangle QAB \sim \triangle QCD$.
- (iii) $\Leftrightarrow$ (iv): This is obvious.

Combining all these, all items are equivalent. $\square$

In other words, [proposition 3.4](#) gives certain ways to prove that four points are concyclic and provides useful facts about concyclic points. Note that the results are not limited to those listed in the proposition as there are many similar results. For example, the equal angles in (ii) correspond to four pairs of angles as shown.

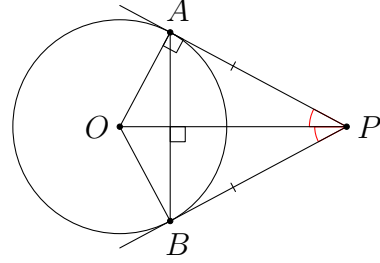


Definition 3.2. A straight line is called a *tangent* (切線) to a circle if it has exactly one intersection with the circle.

Consider a circle and a point P . If P lies on the circle, then there exists a unique tangent to the circle passing through P . If P lies outside the circle, then there exist two tangents passing through P . If P lies inside the circle, there is no tangent passing through P .

Proposition 3.5. Let P be a point outside a circle with centre O . Let PA and PB be the tangents passing through P with contact points A and B . Then the following hold.

- (a) $OA \perp PA$
- (b) $PA = PB$
- (c) $\angle APO = \angle BPO$
- (d) $AB \perp PO$



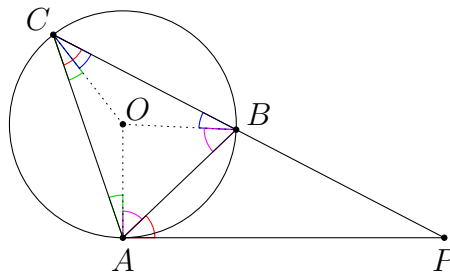
Proof. Let A' be the projection of O on PA , and let A'' be the symmetric point of A with respect to A' . Then $OA = OA''$ by symmetry, which shows that A'' is another intersection point of PA and the circle. By the definition of tangents, we must have $A = A''$, and hence $A = A'$. This proves (a).

By symmetry, we have $\angle OBP = 90^\circ = \angle OAP$. Also, we have $OA = OB$ and $OP = OP$. Thus, we get $\triangle OAP \cong \triangle OBP$ (RHS). (b) and (c) are corollaries of this.

Lastly, (d) follows from (b) and (c) since this is a property of isosceles triangles. $\square$

Proposition 3.6. Let A, B, C be three points on a circle. Let P be the intersection of the tangent at A and the line BC as shown. Then the following hold.

- (a) $\angle PAB = \angle ACP$
- (b) $PA^2 = PB \times PC$



Proof.

- (a) Let O be the centre of the circle. We only consider the case when O lies inside $\triangle ABC$. Let $\angle OBC = \angle OCB = \alpha$, $\angle OCA = \angle OAC = \beta$, $\angle OAB = \angle OBA = \gamma$. Note that $2\alpha + 2\beta + 2\gamma = 180^\circ$ by considering the sum of internal angles of $\triangle ABC$. By proposition 3.5(a), we have

$$\angle PAB = 90^\circ - \gamma = \alpha + \beta = \angle ACP.$$

(b) By part (a), we have $\triangle PAB \sim \triangle PCA$. This implies $\frac{PA}{PC} = \frac{PB}{PA}$. The result then follows.

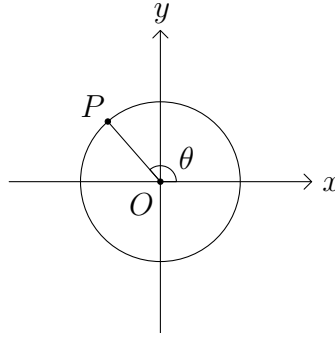
□

Conversely, each of the results in proposition 3.6 implies PA is tangent to the circle passing through A, B, C .

4 Trigonometry

Consider the coordinate plane with origin O . For each angle θ , let $P(a, b)$ be the point on the unit circle such that the directed angle measured anticlockwise from the positive x -axis to the ray OP is θ . Then we define three commonly used *trigonometric functions* (三角函数) as follows.

- *sine function* (正弦函数): $\sin \theta = b$
- *cosine function* (餘弦函数): $\cos \theta = a$
- *tangent function* (正切函数): $\tan \theta = \frac{b}{a}$ (which is only well-defined when $a \neq 0$)



Proposition 4.1. Let θ be any angle measured in degree. Then the following hold.

- (a) $\sin(\theta + 360^\circ) = \sin \theta$
- (b) $-1 \leq \sin \theta \leq 1$
- (c) $0 < \sin \theta \leq 1$ for $0^\circ < \theta < 180^\circ$, increasing on $0^\circ < \theta < 90^\circ$ and decreasing on $90^\circ < \theta < 180^\circ$
- (d) $\sin \theta = \sin(180^\circ - \theta)$ and $\sin(-\theta) = -\sin \theta$

Also, the following table gives the values of $\sin \theta$ for some common angles θ .

θ	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0

As a corollary, if x and y satisfy $0^\circ < x, y < 180^\circ$ and $\sin x = \sin y$, then we have $x = y$ or $x = 180^\circ - y$.

Proposition 4.2. Let θ be any angle measured in degree. Then the following hold.

- (a) $\cos(\theta + 360^\circ) = \cos \theta$
- (b) $-1 \leq \cos \theta \leq 1$
- (c) $-1 < \cos \theta < 1$ and decreasing on $0^\circ < \theta < 180^\circ$
- (d) $\cos \theta = -\cos(180^\circ - \theta)$ and $\cos(-\theta) = \cos \theta$
- (e) $\cos \theta = \sin(90^\circ - \theta)$ and $\sin \theta = \cos(90^\circ - \theta)$
- (f) $\sin^2 \theta + \cos^2 \theta = 1$

Also, the following table gives the values of $\cos \theta$ for some common angles θ .

θ	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1

Proposition 4.3. Let θ be any angle measured in degree. Then the following hold.

- (a) $\tan(\theta + 180^\circ) = \tan \theta$
- (b) $\tan 90^\circ$ is undefined
- (c) $\tan \theta$ increases from 0 to ∞ on $0^\circ < \theta < 90^\circ$, and increases from $-\infty$ to 0 on $90^\circ < \theta < 180^\circ$
- (d) $\tan \theta = -\tan(180^\circ - \theta)$ and $\tan(-\theta) = -\tan \theta$
- (e) $\tan \theta = \frac{\sin \theta}{\cos \theta}$

Also, the following table gives the values of $\tan \theta$ for some common angles θ .

θ	0°	30°	45°	60°	120°	135°	150°	180°
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Proposition 4.4. (Compound angle formulae(和差公式)) Let x and y be any angles. Then the following hold.

- (a) $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
- (b) $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
- (c) $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$

Proposition 4.5. (Double angle formulae(倍角公式)) Let x be any angle. Then the following hold.

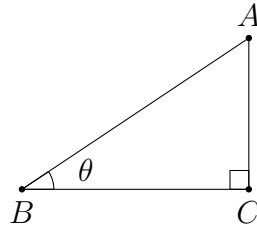
- (a) $\sin 2x = 2 \sin x \cos x$
- (b) $\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$
- (c) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

Proposition 4.6. (Half angle formulae(半角公式)) Let x be any angle. Then the following hold.

- (a) $\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$
- (b) $\cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}$

Consider a right-angled $\triangle ABC$ with $\angle C = 90^\circ$ and $\angle B = \theta$. By definition and by similar triangles, we have

$$\sin \theta = \frac{AC}{AB}, \quad \cos \theta = \frac{BC}{AB}, \quad \tan \theta = \frac{AC}{BC}.$$



These relations allow us to express ratios of side lengths as the trigonometric functions. Some more important results are given below. For the notations used, one may refer to section 5.

Theorem 4.1. (Sine law(正弦定理)) Let $\triangle ABC$ be any triangle. Then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Proof. See theorem 2.1 of Calculating Geometry 3.0. □

Theorem 4.2. (Cosine formula(餘弦定理)) Let $\triangle ABC$ be any triangle. Then

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Equivalently,

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Proof. See theorem 2.2 of Calculating Geometry 3.0. □

As a special case, we have the following theorem.

Theorem 4.3. (Pythagoras' theorem(畢氏定理)) Let $\triangle ABC$ be any triangle. Then $C = 90^\circ$ if and only if $c^2 = a^2 + b^2$.

5 Triangle Geometry

First of all, we list out the common notations used in triangle geometry (some terminologies will be defined in subsection 5.2). But it is a good habit to define everything not involved in the question whenever we use. Let $\triangle ABC$ be a triangle.

- a, b, c : the side lengths BC, CA, AB respectively
- s : the *semiperimeter*(半周長) $\frac{a+b+c}{2}$
- $[ABC]$: the area of $\triangle ABC$
- R : the circumradius of $\triangle ABC$
- r : the inradius of $\triangle ABC$
- G : the centroid of $\triangle ABC$
- O : the circumcentre of $\triangle ABC$
- H : the orthocentre of $\triangle ABC$
- I : the incentre of $\triangle ABC$

5.1 General Facts

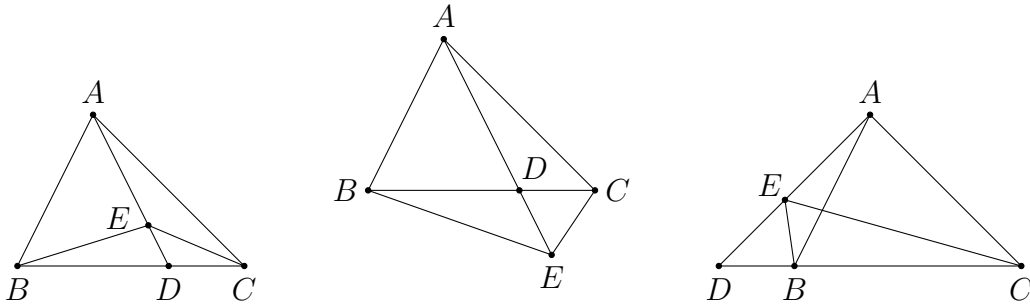
The following are some elementary results related to triangle geometry.

Proposition 5.1. In $\triangle ABC$, let D be a point on the line BC , and let E be a point on the line AD . Then

$$\frac{[ABE]}{[ACE]} = \frac{BD}{CD}.$$

In particular, we have

$$\frac{[ABD]}{[ACD]} = \frac{BD}{CD}.$$



The above figures give a few possible configurations. But there are far a lot more different configurations in which the proposition is applicable.

Proof. See [proposition 2.1](#) of **Calculating Geometry 3.0**. □

Proposition 5.2. Let $\triangle ABC$ be any triangle. Then

$$[ABC] = \frac{1}{2}ab \sin C = \frac{abc}{4R} = sr = \sqrt{s(s-a)(s-b)(s-c)}.$$

Proof. See [proposition 2.2](#) of **Calculating Geometry 3.0**. □

The formula $[ABC] = \sqrt{s(s-a)(s-b)(s-c)}$ is called [Heron's formula](#)(海倫公式).

Example 5.1. In $\triangle ABC$, $BC = 13$, $CA = 14$ and $AB = 15$. The semiperimeter is equal to $s = \frac{13 + 14 + 15}{2} = 21$. By Heron's formula, we have

$$[ABC] = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(21)(8)(7)(6)} = 84.$$

By [proposition 5.2](#), we find that

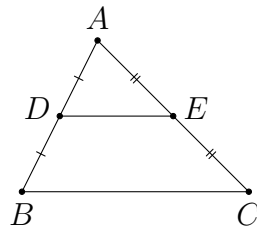
$$R = \frac{abc}{4[ABC]} = \frac{(13)(14)(15)}{4(84)} = \frac{65}{8},$$

$$r = \frac{[ABC]}{s} = \frac{84}{21} = 4.$$

Theorem 5.1. ([Midpoint theorem](#)(中點定理)) In $\triangle ABC$, let D and E be the mid-points of the sides AB and AC respectively. Then the following hold.

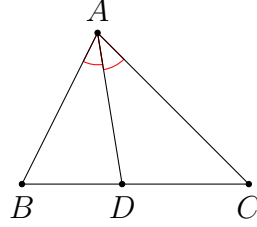
(a) $DE \parallel BC$

(b) $DE = \frac{1}{2}BC$



Proof. By the intercept theorem, we have $DE \parallel BC$ as $\frac{AD}{DB} = \frac{AE}{EC} = 1$. In addition, since $\triangle DAE \sim \triangle BAC$, we get $\frac{DE}{BC} = \frac{AD}{AB} = \frac{1}{2}$. This implies $DE = \frac{1}{2}BC$. □

Theorem 5.2. (Angle bisector theorem(角平分線定理)) In $\triangle ABC$, let D be a point on the side BC . Then $\angle BAD = \angle CAD$ if and only if $\frac{AB}{AC} = \frac{BD}{CD}$.



Proof. By proposition 5.1, we have $\frac{[ABD]}{[ACD]} = \frac{BD}{CD}$. Using proposition 5.2, we find that

$$\begin{aligned} [ABD] &= \frac{1}{2} AB \times AD \sin \angle BAD, \\ [ACD] &= \frac{1}{2} AC \times AD \sin \angle CAD. \end{aligned}$$

This implies

$$\frac{BD}{CD} = \frac{AB \sin \angle BAD}{AC \sin \angle CAD}.$$

Thus, $\frac{AB}{AC} = \frac{BD}{CD}$ if and only if $\sin \angle BAD = \sin \angle CAD$. Now,

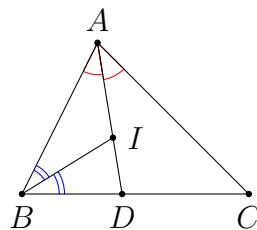
$$\sin \angle BAD = \sin \angle CAD \Leftrightarrow \angle BAD = \angle CAD \quad \text{or} \quad \angle BAD + \angle CAD = 180^\circ$$

as both angles lie between 0° and 180° . However, as $\angle BAD + \angle CAD = \angle BAC < 180^\circ$, the second case cannot happen. This establishes the theorem. $\square$

Example 5.2. In $\triangle ABC$, let D be the foot of internal angle bisector from $\angle A$. The internal angle bisector from $\angle B$ meets AD at I (indeed, I is the incentre of $\triangle ABC$). By the angle bisector theorem, we have $\frac{BD}{CD} = \frac{c}{b}$. Since $BD + CD = a$, we easily obtain

$$BD = \frac{ac}{b+c}, \quad CD = \frac{ab}{b+c}.$$

Also, by the angle bisector theorem, we have $\frac{AI}{ID} = \frac{AB}{BD} = \frac{b+c}{a}$.

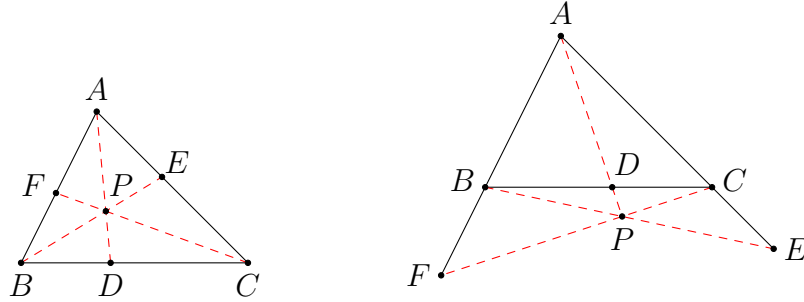


Definition 5.1. Three lines are said to be *concurrent*(共點) if they intersect at a single point.

Definition 5.2. Three points are said to be *collinear*(共線) if they lie on a single line.

Theorem 5.3. (*Ceva's theorem*(塞瓦定理)) In $\triangle ABC$, let D, E, F be points on the sidelines BC, CA, AB respectively. Then AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1.$$

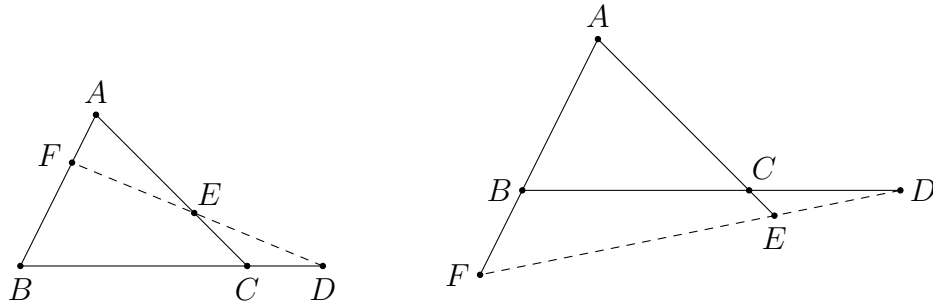


The lengths used in this theorem are directed lengths. Equivalently, one can only apply Ceva's theorem when 0 or 2 of the points D, E, F lie outside $\triangle ABC$, like the configurations shown above.

Proof. See [theorem 3.1](#) of **Calculating Geometry 3.0**. □

Theorem 5.4. (*Menelaus' theorem*(梅涅勞斯定理)) In $\triangle ABC$, let D, E, F be points on the sidelines BC, CA, AB respectively. Then D, E, F are collinear if and only if

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = -1.$$



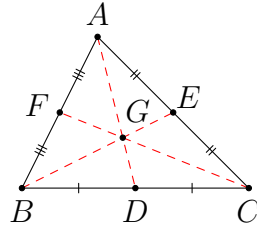
Again, the lengths are directed lengths. Equivalently, one can only apply Menelaus' theorem when 1 or 3 of the points D, E, F lie outside $\triangle ABC$, like the configurations shown above. In that case, the constant -1 can be regarded as 1 .

Proof. See [theorem 3.2](#) of **Calculating Geometry 3.0**. □

5.2 Triangle Centres

Definition 5.3. In $\triangle ABC$, let D, E, F be the midpoints of the sides BC, CA, AB respectively. Then the *medians* (中線) AD, BE, CF are concurrent at a point G , called the *centroid* (重心) of $\triangle ABC$.

Proof. Since $\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1 \times 1 \times 1 = 1$, the three medians are concurrent by Ceva's theorem. □



Proposition 5.3. In $\triangle ABC$, let D, E, F be the midpoints of the sides BC, CA, AB respectively, and let G be the centroid of $\triangle ABC$. Then the following hold.

- (a) $AG : GD = BG : GE = CG : GF = 2 : 1$
- (b) $[AGE] = [AGF] = [BGF] = [BGD] = [CGD] = [CGE] = \frac{1}{6}[ABC]$
- (c) ([Apollonius' theorem](#) (阿波羅尼斯定理)) $AD = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Proof.

- (a) Applying Menelaus' theorem to $\triangle ABD$ and line CGF , we obtain

$$\frac{AG}{GD} \times \frac{DC}{CB} \times \frac{BF}{FA} = 1.$$

Since $\frac{DC}{CB} = \frac{1}{2}$ and $\frac{BF}{FA} = 1$, this yields $AG : GD = 2 : 1$.

(b) By [proposition 5.1](#), we have $\frac{[BGD]}{[CGD]} = \frac{BD}{CD} = 1$, and so $[BGD] = [CGD]$.

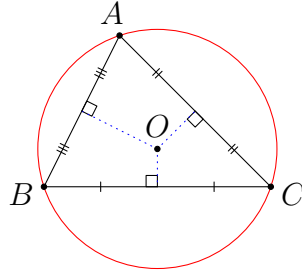
Similarly, we have $[AGF] = [BGF]$. Also, again by [proposition 5.1](#), we have $\frac{[BGC]}{[BGA]} = \frac{CE}{AE} = 1$. This yields $2[BGD] = 2[BGF]$, and hence $[BGD] = [BGF]$. The other equalities hold by symmetry.

(c) Applying the cosine formula to $\triangle ABD$ and $\triangle ABC$, we obtain

$$\begin{aligned} AD^2 &= AB^2 + BD^2 - 2AB \times BD \cos \angle ABD \\ &= c^2 + \frac{a^2}{4} - ac \cdot \frac{a^2 + c^2 - b^2}{2ac} = \frac{1}{4}(2b^2 + 2c^2 - a^2) \end{aligned}$$

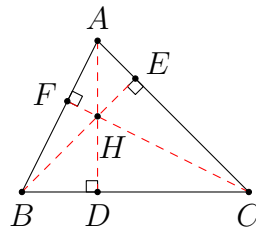
The result follows readily. □

Definition 5.4. In $\triangle ABC$, there is a unique circle, called the [circumcircle](#)(外接圆) and denoted by (ABC) , which passes through A, B, C . The centre O of this circle is called the [circumcentre](#)(外心) of $\triangle ABC$, and the radius R of this circle is called the [circumradius](#)(外接圆半径).



This result is essentially [proposition 3.1](#). In particular, recall O is the intersection of the perpendicular bisectors of BC, CA, AB . All other properties of the circumcentre and the circumcircle come from the properties of circles.

Definition 5.5. In $\triangle ABC$, let D, E, F be the projection from A, B, C to the sides BC, CA, AB respectively. Then the [altitudes](#)(高) AD, BE, CF are concurrent at a point H , called the [orthocentre](#)(垂心) of $\triangle ABC$.



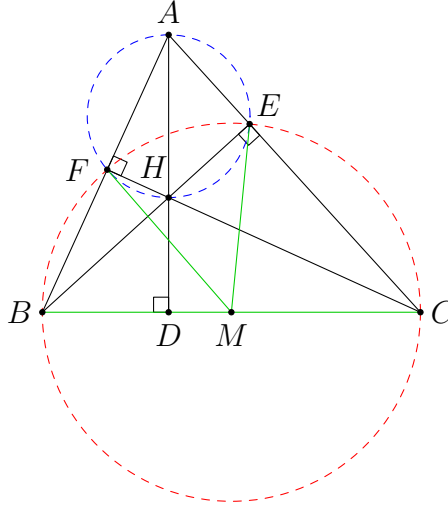
Proof. Using trigonometry, we get $BD = c \cos B$ and $DC = b \cos C$, etc. It follows that

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = \frac{c \cos B}{b \cos C} \times \frac{a \cos C}{c \cos A} \times \frac{b \cos A}{a \cos B} = 1.$$

Thus, AD, BE, CF are concurrent by Ceva's theorem. $\square$

Proposition 5.4. In acute $\triangle ABC$, let D, E, F be the feet of altitudes from A, B, C respectively, and let H be the orthocentre of $\triangle ABC$. Then the following hold.

- (a) B, C, E, F are concyclic
- (b) A, F, H, E are concyclic
- (c) $\angle BHC = 180^\circ - A$



Proof.

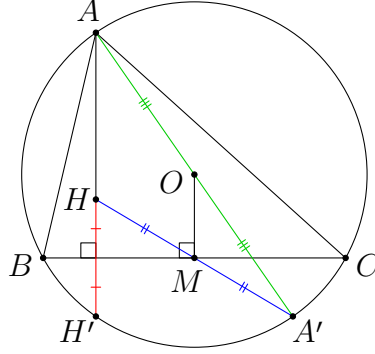
- (a) This is because $\angle BFC = 90^\circ = \angle BEC$.
- (b) This is because $\angle AFH + \angle AEH = 90^\circ + 90^\circ = 180^\circ$.
- (c) This is because $\angle BHC = \angle EHF = 180^\circ - \angle EAF$ in view of the concyclic points A, F, H, E .

$\square$

Note that the centre of the circle passing through B, C, E, F is exactly the midpoint M of BC since $\angle BFC = 90^\circ$. This implies $ME = MF = \frac{a}{2}$. Similarly, the centre of the circle passing through A, F, H, E is the midpoint of AH .

Proposition 5.5. In $\triangle ABC$, let O and H be the circumcentre and orthocentre respectively, and let M be the midpoint of BC . Then the following hold.

- (a) the reflection of H in BC lies on the circumcircle of $\triangle ABC$
- (b) the reflection of H in M coincides with the reflection of A in O (hence also lies on the circumcircle of $\triangle ABC$)



Proof.

- (a) Let H' be the reflection of H in BC . For acute triangles, by the reflection and [proposition 5.4\(c\)](#), we have

$$\angle BH'C = \angle BHC = 180^\circ - \angle BAC.$$

This implies A, B, H', C are concyclic as desired. Other cases can be proved similarly.

- (b) Let A' be the reflection of H in M . Note that $HBA'C$ is a parallelogram since M is the midpoint of both BC and HA' (see [proposition 6.2](#)). Therefore, for acute triangles, we have

$$\angle BA'C = \angle BHC = 180^\circ - \angle BAC.$$

This implies A, B, A', C are concyclic. Other cases can be proved similarly. Also, since $A'C \parallel BH \perp AC$, we have $\angle A'CA = 90^\circ$. This shows AA' is a diameter of the circumcircle of $\triangle ABC$, and hence O is the midpoint of AA' .

□

As a corollary of part (b), we know that $\triangle A'OM \sim \triangle A'AH$, and hence

$$AH = 2OM = 2R \cos A.$$

The last equality holds by considering $\triangle OBM$.

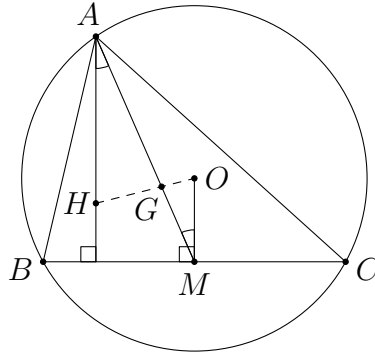
Theorem 5.5. (Euler line(歐拉線)) In $\triangle ABC$, let H, G, O be the orthocentre, centroid and circumcentre respectively. Then H, G, O are collinear in that order with $HG : GO = 2 : 1$.

Proof. Let M be the midpoint of BC . Note that AGM is a straight line, and $AH \parallel OM$. Thus, we have $\angle GAH = \angle GMO$. Also, recall that $\frac{AH}{MO} = 2$. This is equal to $\frac{AG}{MG}$ by proposition 5.3(a). Therefore, we find that $\triangle GAH \sim \triangle GMO$.

It follows that $\angle AGH = \angle MGO$, and hence H, G, O are collinear. Also, we have

$$\frac{HG}{OG} = \frac{AG}{MG} = 2.$$

□



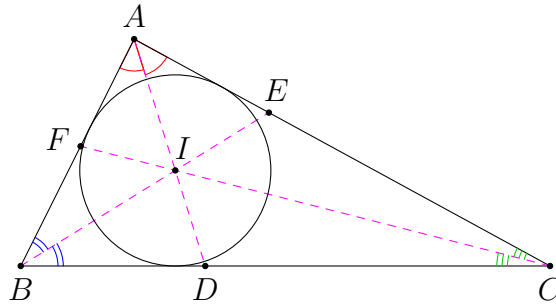
Definition 5.6. In $\triangle ABC$, the internal angle bisectors of $\angle A, \angle B, \angle C$ are concurrent at a point I , called the *incentre*(內心) of $\triangle ABC$.

Proof. Let D, E, F be the feet of internal angle bisectors from A, B, C respectively. By the angle bisector theorem, we have $\frac{BD}{DC} = \frac{AB}{AC}$. Similarly, we have $\frac{CE}{EA} = \frac{BC}{BA}$ and $\frac{AF}{FB} = \frac{CA}{CB}$. Then

$$\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = \frac{AB}{AC} \times \frac{BC}{BA} \times \frac{CA}{CB} = 1.$$

Thus, AD, BE, CF are concurrent by Ceva's theorem.

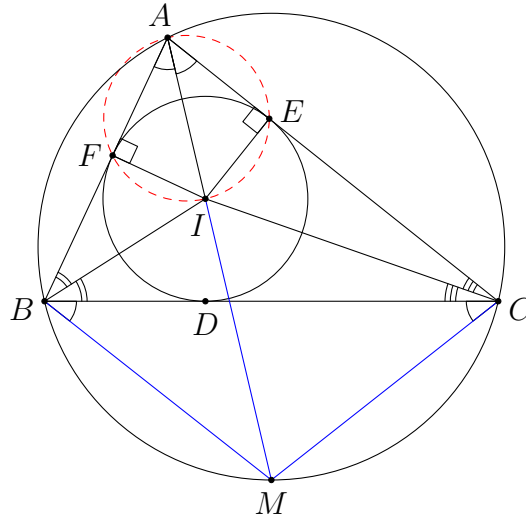
□



Note that any point on the internal angle bisector of $\angle A$ is equidistant to AB and AC . Therefore, the point I is equidistant to AB, BC, CA . Therefore, we can draw a circle with centre I touching the sides BC, CA, AB . This circle is called the *incircle* (内切圓) of $\triangle ABC$, and its radius r is called the *inradius* (内切圓半徑). Indeed, there is exactly one circle inside $\triangle ABC$ which is tangent to all the sides.

Proposition 5.6. In $\triangle ABC$, let O and I be the circumcentre and incentre, and let D, E, F be the contact points of the incircle with the sides BC, CA, AB respectively. Also, let M be the midpoint of $\widehat{BC}$ such that A and M lie on different sides of BC . Then the following hold.

- (a) $\angle BIC = 90^\circ + \frac{1}{2}A$
- (b) $AE = AF = s - a$
- (c) A, F, I, E are concyclic
- (d) $MB = MI = MC$
- (e) (Euler's formula (歐拉公式)) $OI^2 = R^2 - 2Rr$



Proof.

(a) We have $\angle BIC = 180^\circ - \angle IBC - \angle ICB = 180^\circ - \frac{B}{2} - \frac{C}{2} = 90^\circ + \frac{A}{2}$.

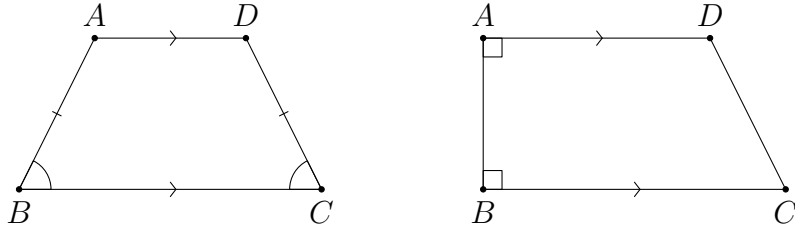
(b) Let $AE = AF = x$, $BF = BD = y$ and $CD = CE = z$. Since $x + y = c$, $y + z = a$ and $z + x = b$, we have

$$x = \frac{(x + y) + (z + x) - (y + z)}{2} = \frac{c + b - a}{2} = s - a.$$

6 Quadrilaterals

Definition 6.1. A *trapezoid*(梯形) is a convex quadrilateral $ABCD$ with $AD \parallel BC$ (or $AB \parallel DC$).

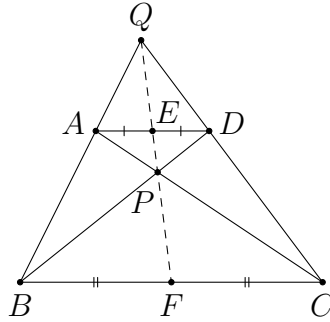
If a trapezoid $ABCD$ with $AD \parallel BC$ satisfies $\angle ABC = \angle DCB$, then it is called an *isosceles trapezoid*(等腰梯形). In that case, we must have $AB = DC$. If a trapezoid $ABCD$ with $AD \parallel BC$ satisfies $A = B = 90^\circ$ (or $C = D = 90^\circ$), then it is called a *right-angled trapezoid*(直角梯形).



Proposition 6.1. A trapezoid is cyclic if and only if it is an isosceles trapezoid.

Proof. Consider a trapezoid $ABCD$ with $AD \parallel BC$. It is a cyclic quadrilateral if and only if $A + C = 180^\circ$. Since $A + B = 180^\circ$, this proves $ABCD$ is cyclic if and only if $B = C$. $\square$

Theorem 6.1. (*Steiner's theorem*(斯坦纳定理)) Let $ABCD$ be a trapezoid with $AD \parallel BC$. Let E and F be the midpoints of the sides AD and BC respectively. Suppose the diagonals AC and BD meet at P , while the lines AB and CD meet at Q . Then Q, E, P, F are collinear.



Proof. By the intercept theorem, we have $\frac{QA}{AB} = \frac{QD}{DC}$. Thus, we have

$$\frac{BF}{FC} \times \frac{CD}{DQ} \times \frac{QA}{AB} = 1 \times \frac{CD}{DQ} \times \frac{QD}{DC} = 1.$$

This proves Q, P, F are collinear by Ceva's theorem. Similarly, we have

$$\frac{AE}{ED} \times \frac{DC}{CQ} \times \frac{QB}{BA} = 1 \times \frac{DC}{CQ} \times \frac{QC}{CD} = 1.$$

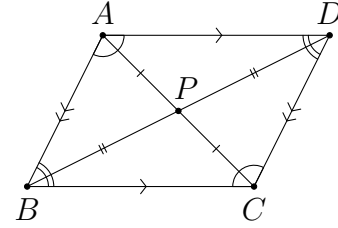
This proves Q, E, P are collinear by Ceva's theorem. Thus, Q, E, P, F are collinear. $\square$

Definition 6.2. A *parallelogram*(平行四邊形) is a convex quadrilateral $ABCD$ for which $AD \parallel BC$ and $AB \parallel DC$.

In particular, a parallelogram is a trapezoid.

Proposition 6.2. Let $ABCD$ be a convex quadrilateral. Let P be the intersection point of the diagonals AC and BD . Then the following are equivalent.

- (i) $ABCD$ is a parallelogram
- (ii) $A = C$ and $B = D$
- (iii) $AD = BC$ and $AB = DC$
- (iv) $AD = BC$ and $AD \parallel BC$
- (v) $PA = PC$ and $PB = PD$



These are fundamental properties of parallelograms. It suffices to notice the congruent triangles $\triangle PAB \cong \triangle PCD$ and $\triangle ABD \cong \triangle CDB$, etc.

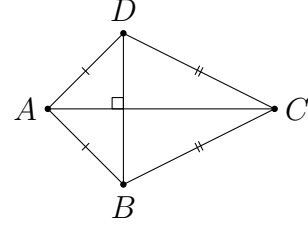
Definition 6.3. A *rectangle*(長方形) is a convex quadrilateral satisfying the condition $A = B = C = D = 90^\circ$.

In particular, a rectangle is a parallelogram. A parallelogram is cyclic if and only if it is a rectangle.

Definition 6.4. A *kite*(鷓鴣形) is a convex quadrilateral for which $AB = AD$ and $CB = CD$.

Proposition 6.3. Let $ABCD$ be a kite with $AB = AD$ and $CB = CD$. Then the following hold.

- (a) $\triangle ABC \cong \triangle ADC$
- (b) $AC \perp BD$



Proof.

- (a) This holds by SSS.
- (b) This is because AC is the perpendicular bisector of BD .

□

Definition 6.5. A *rhombus*(菱形) is a convex quadrilateral satisfying the condition $AB = BC = CD = DA$.

In particular, a rhombus is a parallelogram and a kite.

Definition 6.6. A *square*(正方形) is a convex quadrilateral satisfying the conditions $A = B = C = D = 90^\circ$ and $AB = BC = CD = DA$.

In particular, a square is a rectangle and a rhombus. A rhombus is cyclic if and only if it is a square.

Links

Theorems

- [Angle bisector theorem](#)
- [Apollonius' theorem](#)
- [Ceva's theorem](#)
- [Compound angle formulae](#)
- [Cosine formula](#)
- [Double angle formulae](#)
- [Euler line](#)
- [Euler's formula](#)
- [Euler's inequality](#)
- [Half angle formulae](#)
- [Heron's formula](#)
- [Intercept theorem](#)
- [Menelaus' theorem](#)
- [Midpoint theorem](#)
- [Pythagoras' theorem](#)
- [Sine law](#)
- [Steiner's theorem](#)

Propositions

- [1.1: Angle at a point](#)
- [1.2: Angle sum of polygons](#)
- [1.3: Exterior angle of triangle](#)
- [1.4: Angles and parallel lines](#)
- [1.5: Criterion for a triangle to be isosceles](#)
- [2.1: Criteria for two triangles to be congruent](#)
- [2.2: Criteria for two triangles to be similar](#)
- [3.1: Uniqueness of circle passing through three non-collinear points](#)
- [3.2: Properties of centre of circle](#)
- [3.3: Relations between angles, lengths of arcs and chords](#)

- 3.4: Criteria for four points to be concyclic
- 3.5: Properties of tangents of circle 1
- 3.6: Properties of tangents of circle 2
- 4.1: Properties of the sine function
- 4.2: Properties of the cosine function
- 4.3: Properties of the tangent function
- 4.4: Compound angle formulae
- 4.5: Double angle formulae
- 4.6: Half angle formulae
- 5.1: Ratio of areas
- 5.2: Formulae for the area of triangles
- 5.3: Properties of centroid and medians
- 5.4: Properties of orthocentre and feet of altitudes
- 5.5: Properties of orthocentre and circumcircle
- 5.6: Properties of incentre
- 6.1: Property of isosceles trapezoids
- 6.2: Criteria for a quadrilateral to be a parallelogram
- 6.3: Properties of kites

Terminologies and Notations

- a, b, c
- $[ABC]$
- acute triangle
- altitude
- centroid G
- circumcentre O
- circumcircle
- circumradius R
- collinear
- concave polygon
- concurrent
- concyclic

- congruent triangles $\triangle ABC \cong \triangle XYZ$
- convex polygon
- cosine function $\cos \theta$
- cyclic quadrilateral
- equilateral triangle
- incentre I
- incircle
- inradius r
- isosceles trapezoid
- isosceles triangle
- kite
- median
- obtuse triangle
- orthocentre H
- parallelogram
- rectangle
- rhombus
- right-angled trapezoid
- right-angled triangle
- scalene triangle
- semiperimeter s
- similar triangles $\triangle ABC \sim \triangle XYZ$
- sine function $\sin \theta$
- square
- tangent (circle)
- tangent function $\tan \theta$
- trapezoid
- trigonometric function